

The MindfulMath Tutor: A Mindfulness-Infused Learning Environment to Battle Math Anxiety

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Abstract

Math anxiety is a widespread issue that affects students' emotional well-being and math learning. We developed the MindfulMath Tutor, an interactive mindfulness-infused math learning system with a pedagogical agent that provides both cognitive and socio-emotional support, using positive mindful language and breathing exercises. Our experiment with 31 secondary school students (with medium to high math anxiety) showed that the MindfulMath Tutor helped students significantly reduce their state math anxiety while achieving a similar level of learning benefits, when compared against a tutor version without mindfulness components. Our mindfulness-infused math tutor offers an interactive math learning environment that effectively supports student learning while reducing math anxiety.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**.

Keywords

Math Anxiety, Mindfulness, Emotion Regulation, Educational Technologies

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1 Introduction

Experiencing anxiety in math-related contexts is a widespread issue that affects and impairs students [16, 22, 45, 46], causing them to experience distress during math learning [14, 51]. Math-anxious students may even experience anxiety when just thinking about a situation that involves math [48] and the experience can be as extreme as physical pain-like sensations [36]. Math anxiety has a profound impact not only on students' emotional well-being but also on math learning, as it impairs cognitive functioning when engaging in math tasks [1, 2]. Consequently, math-anxious students often find themselves in a cycle of increased self-doubt and math anxiety,

avoidance of math practice, and poor math performance [2, 36]. Breaking this cycle is crucial in supporting students' math learning and well-being. A way to achieve this is for students to learn to recognize and adaptively regulate their emotions in math-related contexts, for example, through the practice of mindfulness. Mindfulness practice teaches individuals to cultivate conscious awareness of their cognitive processes and bodily sensations [10, 31]. Research has shown that mindfulness-based interventions (e.g., breathing exercises and meditation) can effectively reduce math anxiety and increase performance [11, 20, 35, 53, 58]. However, these interventions focus on mindful activities (e.g., pre-lesson meditation) that often leave students to independently transfer mindfulness skills to high-anxiety math contexts. Also, implementing non-digital mindfulness interventions in educational settings remains challenging due to teachers' existing curricular workload and the absence of mindfulness training in standard teacher training curricula [9].

To overcome these obstacles, we developed a digital mindfulness-infused learning environment that is targeted to seamlessly support learners in regulating negative emotions during math problem solving while supporting math learning – the *MindfulMath Tutor*. It integrates a pedagogical agent that provides cognitive and socio-emotional support, uses mindful language, and offers mindful breathing exercises. This paper outlines the design, development and evaluation of the *MindfulMath Tutor*.

Our online experiment with 31 math anxious school students showed that the MindfulMath Tutor outperformed the Cognitive-Math Tutor, a version of the same learning software without socio-emotional mindful support, in reducing math anxiety while achieving similar levels of learning gains. This work contributes a novel interactive approach that integrates mindfulness support during math learning, supporting both the learning process and math anxiety regulation. Our findings have important implications for math education, highlighting the potential of mindfulness support in learning environments.

2 Theoretical Background and Related Work

2.1 Math Anxiety

Math anxiety (MA) is a pervasive state of distress that impairs one's math problem-solving skills [51], affecting school students [45, 46], especially those in grade six to ten [16, 22]. MA is a distinct construct characterized by a fearful response triggered by math-related contexts, and is correlated with general and test anxiety [14]. Although MA does not correlate with intelligence [22, 37], it negatively correlates with **math performance (MP)** [2, 44, 68]. This effect is attributed to limited cognitive resources (i.e., working memory) that are consumed by the experience of anxiety [2, 23].

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To examine MA at granular levels, we distinguish between MA as a *trait* – a relatively stable personal trait of perceiving math-related contexts as threatening, and MA as a *state* – a moment-to-moment situational anxiety experienced in math-related contexts [48, 61]. MA interventions typically address trait MA, though state MA is more strongly linked to reduced MP [14, 48]. In our work, we capture both trait MA and state MA, with a focus on state MA as our target outcome. We further consider two subtypes of state MA: *Executive state MA* manifests during a math-related situation (e.g., a MP test), and *anticipatory state MA* is experienced in anticipation of a math-related situation [65].

2.2 Mindfulness and its Application in Math Learning

Human behavior and thought processes are often automatic and unconscious [6], which can lead to maladaptive thought patterns and negative emotions (e.g., anxiety) [13]. This can be especially problematic in educational contexts, as they hinder learning processes by consuming working memory resources [2, 23]. Mindfulness practice helps to disrupt automatic thought patterns by fostering conscious awareness of cognitive processes and bodily sensations [10, 31]. Mindfulness is defined as an adaptive *moment-to-moment, non-judgmental awareness of things as they are in the present* [30]. It can free cognitive resources and reduce avoidance of emotional stimuli, such as those experienced by math-anxious students [4]. Mindfulness is often cultivated through exercises that focus intentional awareness on the breath – an accessible way to introduce mindfulness to novice practitioners [31], for example in schools [11]. Kabat-Zinn’s *principles of mindfulness* [30] offer a comprehensive framework to cultivate mindfulness in education:

- (1) **Non-Judgment** – recognizing and stepping back from judgmental and evaluative thoughts.
- (2) **Patience** – recognizing the natural unfolding of events, accepting mastery as a gradual process
- (3) **A Beginner’s Mind** – being open and, with a curious mind, viewing each situation as a novel experience.
- (4) **Self-Trust** – trusting one’s intuition, cultivating self-belief.
- (5) **Non-Striving** – emphasizing the learning process instead of outcomes
- (6) **Acceptance** – acknowledging and embracing things as they are in the present moment.
- (7) **Letting go of Thoughts** – disengaging from negative and distracting thoughts.

These principles are in line with OECD’s [47] suggestions that instruction to prevent math anxiety should encourage a non-judgmental, curious, exploratory approach to math, valuing process over outcome (as also recommended by [54]), learning at an individual pace, and encouraging students to trust their own abilities. Studies that integrated mindfulness into math education in multi-week programs (e.g., Mindfulness And Growth Mindset Approach (MAGMA) [53], Mindful Math program [60]) report a significant reduction in MA among participants. Also brief mindfulness (e.g., breathing) exercises have been shown to reduce MA and improve MP [11, 20, 35, 58]. These existing interventions focus on the implementation of mindfulness exercises *before* a math lesson or task. As

such, students need to independently transfer and apply mindfulness skills during math contexts, which can be challenging when high MA impairs cognitive functioning [2, 23]. Scaffolding students before, during and after math engagement to support their mindfulness skills within critical situations could help to regulate MA more effectively.

The developmental period during adolescence is critical for introducing mindfulness principles that help students to internalize adaptive emotion regulation strategies [8, 10, 17, 67]. Thus, efforts have been made to develop programs to cultivate mindfulness in education in general, without subject-specific focus (e.g., [9, 56]). However, traditional mindfulness programs face implementation barriers in educational contexts due to their reliance on instructor training and dedicated curricular time [9, 27, 33]. Prior work on mindfulness programs targeting MA has largely neglected the potential of digital technologies.

2.3 Interactive Learning Environments to Help Reduce Math Anxiety

Interactive learning technologies, such as Intelligent Tutoring Systems (ITSs) are established tools for enhancing learning outcomes and addressing individual needs of students [34]. They simulate key elements of human tutoring and provide tailored instruction, support, and feedback [34, 63]. A number of interactive learning systems have been shown effective in supporting students’ math learning and performance [12, 25, 43, 66] both in controlled experiments and real-world implementation [50]. However, these systems mostly focus on students’ cognitive (and metacognitive) learning and do not take into account students’ emotional state, which is especially important in the case of MA [26].

A promising approach to help students to regulate MA is embedding *pedagogical agents* into a learning environment. Pedagogical agents are a form of virtual agents designed to support student learning by giving feedback, hints, and socio-emotional support within an interactive system. Studies have shown that virtual agents in general can function as social interaction partners [7, 19], evoke emotional reactions [55, 59], trust [64], and rapport [21], and can aid emotion regulation [32, 49]. Similarly, pedagogical agents in learning technologies offer various roles within students’ learning processes (e.g., as an expert, fellow learner) and provide adaptive messages, feedback, and other types of scaffolding depending on the learner’s problem solving and knowledge status. They can improve learning and foster students’ motivation [15, 29, 57], help students avoid maladaptive learning behaviors [5], and teach metacognitive strategies [3].

Implementing mindfulness through pedagogical agents, a concept that has not yet been explored in the existing literature, could help students by providing a component of socio-emotional support and interpersonal emotion regulation through friendly and positive messages [5, 39, 52]. For example, a pedagogical agent which provides socio-emotional support by taking into account student’s emotions, was found to be more effective when paired with a non-digital mindfulness exercise, resulting in higher programming performance and motivation [42].

Standalone or non-digital mindfulness programs require dedicated instructor and curricular time [9]. By contrast, integrating

mindfulness principles and exercises in an interactive learning technology using a pedagogical agent eliminates the need for teacher training and enables: (1) Minimized additional time investment, (2) just-in-time MA interventions during math tasks through embedded scaffolding, and (3) seamless mindfulness skill transfer by contextualizing mindfulness within math problem-solving.

3 Development of the MindfulMath Tutor

This paper presents the *MindfulMath Tutor*, a learning software designed to help learners regulate negative emotions in math-related contexts, which integrates a pedagogical agent that provides both cognitive and socio-emotional support, conveying principles of mindfulness by using mindful language and offering mindful breathing exercises.

3.1 The Tutor Design

The MindfulMath Tutor has been designed according to Mayer's cognitive design model that aims to facilitate learning and reduce cognitive load [40]. In an iterative adaptive user-centered optimization process, we refined the prototype and study design in multiple feedback rounds with two adolescents, leading to improved color schemes, age-appropriate language, navigation and progress tracking functions, and adjusted time management to ensure participant engagement (shorter learning units, reduction of breathing exercise frequency from 6 to 3 times in total).

The tutor was implemented in Articulate Storyline 3 for tablet and PC. It consists of three learning units on linear equations using multiple-choice and drag-and-drop formats with (meta-)cognitive scaffolding such as hints and feedback messages that are provided adaptively based on learner input (see Fig. 1a and 1b). The learning units segment equation-solving into smaller steps, through which learners go at their own pace, with step-by-step guidance [40].

3.2 The Pedagogical Agent: "Maddox"

In the MindfulMath Tutor, learners are guided by a pedagogical agent, a cartoon-style fox with human-like qualities (see Fig. 1). We chose this embodiment based on research showing that cartoon-like animal characters are perceived as friendly and engaging, leading to positive learning outcomes [5, 28, 62].

Unlike common pedagogical agents who focus on (meta-)cognitive aspects only [15], Maddox further provides socio-emotional support, building rapport with the learner and showing empathy. The interaction unfolds in a conversational style and is personalized: Maddox addresses the learner by name and initiates emotional check-ins about each learning unit (e.g., "Before we start, think about how you're feeling right now. Is there any concern about the upcoming task? It's okay to have a range of feelings. Learning is a unique journey for each of us"). Instructions and feedback, provided by Maddox in the MindfulMath Tutor, are not focused on correctness but emphasize the learning process, non-judgment and acceptance, using a positive mindful language (see Section 3.3).

The instructional content and interactions with Maddox are implemented in a chat-like format, with pre-scripted interactions that simulate real-time conversation. Learners can choose between answer options, triggering fitting responses from Maddox, as illustrated in the following example:

MADDOX: *Before we dive in, take a moment to check in with yourself. How are you feeling?*

USER [answer option 1]: *I am good.* MADDOX: *That's great to hear!*

USER [answer option 2]: *Bit exhausted.* MADDOX: *Learning can be tiring! Take a moment to recharge if you need to.*

3.3 Mindful Language

In order to create a mindfulness-infused learning environment, we applied a positive and mindful language framework based on Kabat-Zinn's principles of mindfulness [30] (see Section 2.2). This framework, validated by a licensed mindfulness-specialized psychotherapist, guides the language used in the MindfulMath Tutor, including task introductions, feedback messages, and user answer options. For example, Maddox refers to the learning units as "learning journey" or "exploring" to emphasize patience, non-striving and beginner's mind. Besides mindful task instructions and feedback messages, the MindfulMath Tutor includes brief mindful dialogues that promote the principles of mindfulness. For example, when a learner indicates "I'm a bit nervous", Maddox responds with reassuring messages, such as "It's okay and normal to feel nervous. We'll take things slowly and tackle each step together. And remember, math is a journey, not a race", promoting acceptance, patience, and socio-emotional support.

Table 1 provides examples of the mindfulness principles and corresponding language used in the tutor. We provide further details on OSF¹.

3.4 Mindful Breathing Exercises

In the MindfulMath Tutor, Maddox guides learners through three brief mindful breathing exercises before each learning unit, using a combination of breathing animations, verbal cues ("breathe in/out"), and calm music to support eight cycles of deep breathing with a duration of one minute (see Fig. 1c), inspired by [18]. Learners are free to opt out of the exercises. The MindfulMath Tutor includes instructions to introduce the breathing exercises (e.g., "I will guide you. We will slowly inhale through the nose into our bellies. Then, we briefly pause and slowly exhale through the mouth. You can place your hands onto your belly to feel how it moves as you breathe").

4 Evaluation of the MindfulMath Tutor

To evaluate the effectiveness of the MindfulMath Tutor, we conducted an online experiment, where we compared the MindfulMath Tutor against a learning system without mindfulness aspects, which we call *CognitiveMath Tutor*. The MindfulMath Tutor offers cognitive support (e.g. hints) while using positive mindful language throughout the interactions (see Sec. 3.3). Also, it provides mindful breathing exercises before each learning unit (see Sec. 3.4). In contrast, the CognitiveMath Tutor focusses on cognitive support, using language that does not incorporate mindfulness principles (i.e., standard messages that are commonly seen in ITSs). Due to instructions and dialogues that promote principles of mindfulness and breathing exercises, interactions with Maddox in the MindfulMath Tutor are slightly longer compared to the CognitiveMath Tutor. Otherwise, design and functionality of both tutors are equivalent.

¹https://osf.io/fe9gd/?view_only=0a423f12d8f34008b11160f33702a0a8

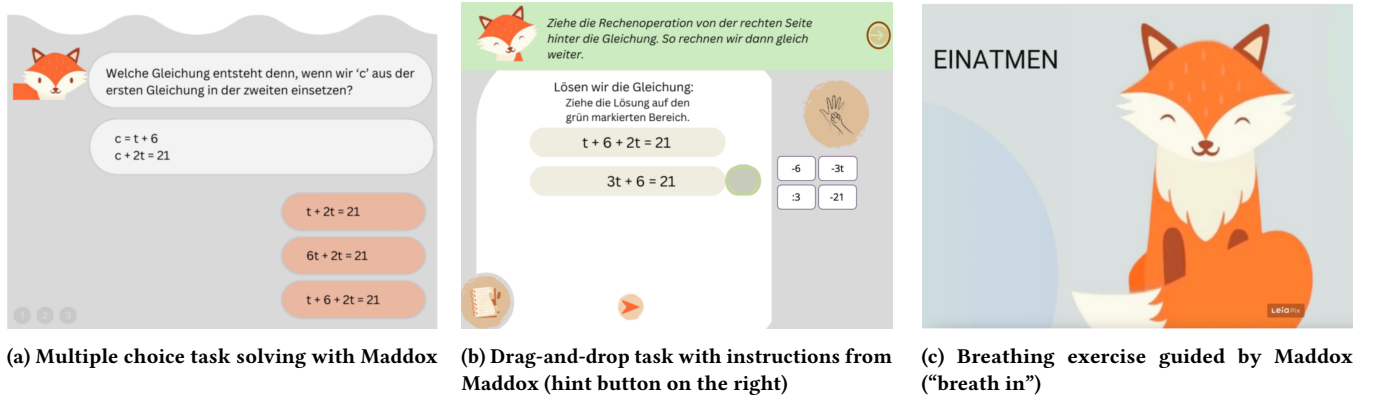


Figure 1: The MindfulMath Tutor Learning Environment with the Pedagogical Agent Maddox (in German Language)

Table 1: The Principles of Mindfulness and their Implementation in the MindfulMath Language

Mindfulness Principle	MindfulMath Language (Examples)
1) Non-Judgment	"That's alright. You're allowed to make mistakes!"
2) Patience	"We're taking this one step at a time, at your own pace."
3) Beginner's Mind	"I'm open to anything that comes next!" (user answer option)
4) Self-Trust	"You can do what feels right for you."
5) Non-Striving	"Let's explore and try out different ways to solve linear equations together."
6) Acceptance	"That's completely fine, uncertainty is normal."
7) Letting go of Thoughts	"Right now, it doesn't matter what happened before or what will happen after this exercise. Let's just focus on what we're doing now."

We investigated the following research question: *How does the MindfulMath Tutor influence students' state math anxiety (MA) and math performance (MP)?* Our hypotheses are: Compared to students learning with the CognitiveMath Tutor, students learning with the MindfulMath Tutor will report a greater reduction in state MA (H1); demonstrate higher pre-post learning gains in terms of MP (H2). In addition, we gather qualitative insights into participants' experiences with the MindfulMath Tutor through follow-up interviews. The study was approved by the Ethical Review Board of an affiliated university.

4.1 Participants

Forty-one Grade 8 and 9 students from German schools participated. Four were excluded due to missing data or withdrawal. The final data set contained 37 participants (64.9% female, 35.1% male; age 13-16, $M = 14.49$, $SD = .73$). Participants were randomly assigned to either the MindfulMath condition or the control (CognitiveMath) condition. They received a compensation of 20 EUR.

4.2 Methods of Data Collection

4.2.1 Math Performance. MP was evaluated using isomorphic pre- and post-tests with six tasks on linear equation systems.

4.2.2 Math Anxiety. **Trait MA** was measured with the German Abbreviated Math Anxiety Scale (AMAS-G) [24], excluding one outdated item. Participants rated their general anxiety in math-related contexts on eight items (e.g., "Taking a test in a math class")

on a 5-point Likert scale (1 = not at all anxious, 5 = very much anxious).

State MA was measured with the six-item version of the state scale of the State-Trait Anxiety Inventory (STAI) [38]. Participants rated their current emotional state (e.g., "I am worried") on a 5-point Likert scale (1 = not at all, 5 = very much). The STAI was administered four times: right before the announced MP pre- and post-test to assess **anticipatory state MA**, and halfway through each MP test to capture **executive state MA** during a math-related situation.

4.2.3 Qualitative Data Collection. We conducted semi-structured chat interviews with all participants in the MindfulMath condition to gain insights into their experiences with the MindfulMath Tutor. Example questions were: *What effect did the breathing exercises have on you? How did you feel about the way Maddox spoke with you? How has Maddox helped you to deal with your feelings while studying?*

4.3 Procedure

Prior to participation, participants and their legal guardian provided informed consent. At an agreed time, participants received an introduction via an *online chat* with the experimenter (via tablet or PC). First, they received a link containing the **pre-test** on MP and questionnaires assessing demographics, trait MA, anticipatory and executive state MA. Then, they received a link to the **learning unit** with the MindfulMath Tutor or the CognitiveMath Tutor. Afterwards, they received a link that contains the **post-test** on

MP and questionnaires assessing anticipatory and executive state MA. This study procedure took on average 45 minutes in both conditions, with slightly longer durations (about 5 to 10 min) in the MindfulMath Tutor condition due to the breathing exercises (see Sec. 3.4), mindful instructions and dialogues (see Sec. 3.3). Participants in the MindfulMath Tutor condition additionally participated in a **semi-structured interview** via the online chat that took approximately 10-15 minutes. The participants completed the entire study procedure without involvement of their legal guardians.

4.4 Analysis

Quantitative analyses were performed using independent samples t-tests and repeated-measure ANOVAs in SPSS with time (pre-test vs. post-test) and condition (CognitiveMath vs. MindfulMath) as factors. Qualitative data was analyzed according to the standardized *qualitative content analysis (QCA)* protocol *inductive category formation* by [41] in the software QCMap², to identify themes in participants' responses. Two independent coders conducted the QCA, with inter-coder agreement assessed, following [41].

5 Results

5.1 Quantitative Analyses

5.1.1 Baseline Measure: Trait Math Anxiety. The mean trait MA was $M = 20.76$ ($SD = 4.99$). Six participants were categorized as low (15.77 and below). Given our focus on math anxious students, we included 31 participants with medium (15.78 – 25.75) and high (25.76 and above) trait MA in the quantitative analyses. We found no significant difference in trait MA between the CognitiveMath ($M = 21.47$, $SD = 3.53$) and MindfulMath conditions ($M = 20.00$, $SD = 6.19$), $t(30) = .88$, $p = .19$, $d = .17$.

5.1.2 State Math Anxiety and Math Performance. Means and standard deviations of state MA scores and MP scores for the CognitiveMath and MindfulMath conditions at pre-test and post-test are shown in Table 2.

To address **H1**, we report anticipatory state MA, executive state MA, and overall state MA (anticipatory and executive MA scores combined). We found that **overall state MA** decreased significantly more in the MindfulMath condition with a significant main effect of time ($F(1, 29) = 16.135$, $p = .001$, $\eta^2 = .357$) and a significant time-by-condition interaction ($F(1, 29) = 6.843$, $p = .014$, $\eta^2 = .191$). When examining the state MA subtypes, we found, for **anticipatory state MA**, no significant main effect for time ($F(1, 29) = 2.71$, $p = .11$, $\eta^2 = .085$) or time-by-condition interaction ($F(1, 29) = 1.14$, $p = .295$, $\eta^2 = .038$), although descriptive analyses show an average decrease of 2.08 in the MindfulMath condition. For **executive state MA**, we found a significant main effect for time ($F(1, 29) = 20.54$, $p < .001$, $\eta^2 = .415$) and a significant time-by-condition interaction ($F(1, 29) = 8.75$, $p = .006$, $\eta^2 = .232$), with a greater decrease in the MindfulMath condition. Addressing **math performance (H2)**, we found a significant main effect for time ($F(1, 29) = 5.80$, $p = .023$, $\eta^2 = .17$), but no significant interaction of time and condition ($F(1, 29) = .04$, $p = .838$, $\eta^2 = .001$), indicating that both tutors had a similarly positive effect on MP.

²www.qcmap.org

5.2 Qualitative Analyses

We report the most notable themes that have emerged in a qualitative content analysis (QCA) of interviews with participants from the MindfulMath condition, with examples of participant quotes. QCA reliability was ensured through inter-coder agreement, reaching excellent agreement rates in QCMap [41].

First, participants viewed the **breathing exercises** introduced in the MindfulMath Tutor as **relaxing and calming** ("I became calmer after the first tasks because of the breathing." – P9, "I liked it because I could relax briefly and collect myself between the exercises." – P5), though some participants preferred to do them **less frequently**. One participant also expressed their intention to apply the breathing exercise in future academic contexts.

Further, participants reported that Maddox provides **positive emotion regulation support**: "If you study with Maddox the night before a test, it can really reduce stress and prevent mental breakdowns." (P10); They labeled Maddox as **kind and empathetic**: "The fox was very nice" (P9); "He always took my feelings into consideration" (P2); "If there would be such a teacher, who would approach new topics so understandingly and calmly with us, I would probably have better grades in math" (P11); and as **supportive and companion-like**: "He was very motivating" (P1); "I think, because of Maddox (...), I just didn't feel so alone with the problem" (P11); "It was somehow really confidential, as if we were friends" (P7). They valued Maddox' **non-judging and accepting approach**: "He often said that it's okay to make mistakes" (P4); "He took away my fear of making mistakes" (P10). While most participants rated Maddox' communication style positively, a few noted that it was **not age appropriate**: "It was a bit over the top and made me feel like a child" (P10); "a bit childish for ninth grade, but otherwise good" (P9). Many participants also gave **general positive feedback**: "It was definitely more fun than in class, and I think it's a great solution for people who struggle with math" (P7); "I think the learning software is a great help with math, because Maddox explains every step quite well and doesn't skip any, so that you can understand everything well. I really liked and enjoyed it" (P1).

6 Discussion and Conclusion

This paper presents the *MindfulMath Tutor*, a mindfulness-infused learning environment with a pedagogical agent that provides both cognitive and socio-emotional support, conveying principles of mindfulness by using mindful language and offering breathing exercises, to help learners regulate negative emotions in math-related contexts.

Our study showed that, in comparison to the CognitiveMath Tutor – an equivalent learning environment without mindfulness aspects – the MindfulMath Tutor helped students to better regulate their **math anxiety (MA)**. Specifically, the MindfulMath Tutor had a substantial effect on *executive state MA*: Students experienced less MA *during* a math performance test, after practicing with the MindfulMath Tutor. On the other hand, students experienced only a slight reduction in anticipatory MA after practicing with the MindfulMath Tutor. That is, the MindfulMath language and breathing exercises seem to be more effective in preparing students to better regulate their MA while working on a math task, but not as effective in addressing future-oriented concern about a math task.

Table 2: Means (with Standard Deviations) of State Math Anxiety and Math Performance by Condition and Time Point

	MindfulMath Tutor		CognitiveMath Tutor	
	Pre-test	Post-test	Pre-test	Post-test
Overall State Math Anxiety	33.31 (8.38)	25.15 (6.36)	29.06 (9.15)	29.33 (8.10)
Anticipatory State Math Anxiety	14.85 (4.98)	12.77 (4.30)	14.44 (4.30)	14.00 (3.68)
Executive State Math Anxiety	18.46 (5.04)	12.38 (2.84)	16.61 (5.93)	15.33 (5.02)
Math Performance	2.77 (0.83)	3.46 (1.49)	2.75 (1.29)	3.33 (1.08)

However, students reported generally lower levels of anticipatory state MA to start with, compared to executive state MA.

Despite the frequently reported negative impact of MA on **math performance (MP)** [2, 44], and while the MindfulMath Tutor helped students reduce their state MA, it did not lead to differential gains in MP. In fact, students using the MindfulMath and CognitiveMath Tutors both improved their MP, indicating that the mindfulness component did not significantly impact math learning. This may be due to an effective design of our learning environment following established instructional design principles, and the use of pedagogical tutor Maddox, which provided clear support throughout the learning process in both tutor versions.

Importantly, the study showed that equipping the pedagogical agent with mindfulness components makes students' experience more meaningful by *reducing MA while keeping learning benefits*, when compared against the version with an agent only providing cognitive support. Consequently, the implementation of mindfulness components in a math learning environment appears to be key to supporting students in regulating MA. Our results align with previous research on MA-reducing mindfulness interventions (e.g., [11, 42, 53]). Yet, previous approaches do not make use of modern learning technologies and rely on self-contained mindfulness exercises that are not seamlessly integrated into the learning process. Our approach is unique in its use of interactive technologies to augment the entire learning process with a mindfulness component, employing the a mindful language that conveys a non-judgmental, process-oriented and patient approach to math [30].

Qualitative interviews showed that the design of our pedagogical agent and the implementation of our MindfulMath language framework, as well as the breathing exercises, were successful in achieving the intended effects. Participants valued Maddox' mindful and emotional support, reducing anxiety and fear of making mistakes, describing it as kind, empathetic, supportive, and non-judging, with some even considering it a companion. They emphasized the MindfulMath Tutor's benefits for students struggling with math. However, in response to concerns expressed by students, we plan to refine the MindfulMath language to better suit the targeted age group. The breathing exercises were rated as calming and relaxing, but our results suggest that students should be given the autonomy to choose when to engage in them.

We acknowledge several limitations of the study. First, we had a small sample size of 37 students. As math anxious students tend to avoid math-related activities [60], recruiting them for a math-related study proved challenging. Despite filtering out students who reported to like math in a pre-screening, an uneven distribution of participants with low MA (5 experimental, 1 control) further limited our sample size for quantitative analyses. Also, due to the

online setup, we cannot rule out the possibility that participants, even if they reported that they had done breathing exercises, might have not actually engaged in the exercises. Future studies could replicate the present work in an in-person lab or classroom environment, with a larger sample size. Further, we aimed to ensure comparability of the setup between the MindfulMath and CognitiveMath. However, we did not track participants' time spent on dialogues with Maddox and math tasks. While the mindful breathing exercises were fixed at one minute, dialogue and task durations were self-paced. Also, interactions in the MindfulMath Tutor took slightly longer due to additional mindful dialogues and instructions. Importantly though, both conditions received the same amount of math problems. Future studies could control and quantify time distribution between mindfulness and math components to allow for stronger conclusions about the effects of mindfulness on MA reduction and MP improvement.

In sum, the MindfulMath Tutor, providing a mindfulness-infused math learning environment, shows great potential in preventing and reducing students' math anxiety. We propose the use of the MindfulMath Tutor as a valuable adjunct to traditional math education in both school and home settings, providing students with a supportive, interactive math learning experience to build a more positive relationship with mathematics.

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